

Samples.csv = metadata file

- Put this into module 4 folder

Promicromonospora.gff - annotated genome

Promicromonospora.fasta - genome in a list

Step 1.) check for quality with fastqc (use falco [Galaxy Version 1.2.4+galaxy](#) because its faster)

- Default parameters fine
- Download forward and reverse quality reads
  - Forward data: fails in Per Tile Sequence Quality, PerBase Sequence Content, Per Sequence GC Content, Sequence Duplication Levels, and Overrepresented Sequences
    - Warning in the Sequence Length Distribution
  - Reverse Data: same fails and warning as forward sequence reads

Step 2.) trim with fastp ([Galaxy Version 1.3.1+galaxy0](#))

- Set to paired collection
- Gets rid of stuff that are fails for the quality control
  - Observe the results on HTML, those under filtering reports

Step 3.) HISAT2 ([Galaxy Version 2.2.2+galaxy0](#)) - aligns transcriptome reads to the genome that is provided by Dr. Van Laar

- Change in parameters:
  - Source for reference genome: genome from the history
  - Single or paired library? Set to paired-end dataset collection

Made an alignment file called BAM

Now, what actual transcripts do we have?

Step4.) StringTie ([Galaxy Version 2.2.3+galaxy0](#)) - assembles short reads into actual transcripts, predicts full length transcripts

- Make sure input option parameter is set to Short Reads

Created 6 files of transcripts

Step 4.) StringTieMerge ([Galaxy Version 2.2.3+galaxy0](#)) - instead of 6 files, puts them all together

- Use dataset collection and use the StringTie file

Step 5.) Gffcompare - takes all assembled transcripts and compare to genome annotation

- Add gff file as part of the history
  - Simply add it to the StringTie merge file, you can do this because it takes multiple datasets

Step 6.) featurecounts - figure out in each samples how many time does each of those samples actually occur, measures gene expression

- Set alignment file as Dataset collection
- Gene annotation file: A GFF/GTF file in your history
- Does input have read pairs? Yes, paired-end and count them as 1 single fragment